

Scalability & Future Plan

Date	29 September 2026
Team ID	SWTID-2026-2825
Project Name	AI WeatherWise
Maximum Marks	1 Mark

Current System Limitations

S.No	Limitation	Impact	Priority to Address
1	No frontend client UI – API-only backend	Harder for non-technical evaluators/users to try the product directly	Medium
2	Single-instance local MongoDB, not yet configured for replica sets	No built-in redundancy if the database instance goes down	Medium
3	Free-tier OpenWeatherMap & Gemini API rate limits	Caps testing/usage throughput during heavy load	High

Scalability Plan

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single local Node.js instance	Deploy stateless instances behind a load balancer (e.g. Render/Railway autoscale) – JWT auth requires no server-side session, so horizontal scaling is straightforward
2	Data Storage	Single local MongoDB instance	Migrate to MongoDB Atlas with replica sets and sharding for the Users/Locations collections
3	Performance	Direct external API call per request	Add a caching layer (e.g. Redis) for frequent city look-ups to cut OpenWeatherMap/Gemini calls
4	Security	JWT + bcrypt + basic sanitization	Add rate-limiting middleware, refresh tokens, and helmet.js security headers

Future Roadmap

The extended data model explored during design (see the ER Diagram in Project Documentation) already outlines the next three phases of growth:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Add an Alerts collection – push/email alerts when a saved city's conditions cross a severe-weather threshold	October 2026	Higher user retention and real safety value
Phase 3	Add Subscriptions & AI_Models collections – tiered plans and a choice of Gemini/other models per insight request	November 2026	Monetization path plus AI model flexibility
Phase 4	Add an API_Usage_Logs collection with an admin analytics dashboard, plus a lightweight frontend client	December 2026	Operational visibility, cost control, and broader usability beyond API clients